Problem Sheet: Sets, Relations, and Functions

Instructions. Show your reasoning clearly. Unless specified otherwise, the universal set for complements is either stated in the problem or is clear from context. Provide counterexamples where a statement is false.

(1) Set Theory

S1: Let $A = \{1, 2, 3, 4, 5\}$, $B = \{3, 4, 5, 6, 7\}$, $C = \{5, 6, 7, 8, 9\}$. Compute:

(a) $(A \cup B) \cap C$

(b) $A \cap (B \cup C)$

(c) $(A \cap B')' \cap C$ assuming $U = \{1, 2, \ldots, 10\}$.

S2: In a group of 100 students, 60 like Mathematics, 50 like Physics, and 30 like both. Find the number who like (i) only Mathematics, (ii) only Physics, (iii) neither subject.

S3: (Optional) Prove or disprove: $A \times (B \cup C) = (A \times B) \cup (A \times C)$ for arbitrary sets A, B, C.

S4: Let $A = x \in \mathbb{R} : x^2 \leq 9$ and $B = x \in \mathbb{R} : -1 < x < 5$. Find $A \cap B$, $A \cup B$, and $A \setminus B$ (describe as intervals).

(2) Relations

R1: Let $A = \{1, 2, 3\}$ and $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}$. Determine whether R is (i) reflexive, (ii) symmetric, (iii) antisymmetric, (iv) transitive.

R2: On $\mathbb{Z}$, define $aRb \iff 4 \mid (a - b)$.

- Show that R is an equivalence relation.
- (Optional) List the equivalence classes of $0, 1, 2, 3$.

R3: Let $A = \{1, 2, 3\}$ and $R = \{(1, 2), (2, 3), (1, 3)\}$. Is R transitive? If not, add the minimum pairs to make it transitive.

R4: Define R on $\mathbb{R}$ by $aRb \iff a^2 + b^2 = 1$. Determine which of: reflexive, symmetric, transitive hold. Justify each.

R5: Let $A = \{1, 2, 3, 4\}$ and $R = \{(a, b) : a \leq b\}$. Identify all properties satisfied among: reflexive, symmetric, antisymmetric, transitive. Briefly justify.

(3) Functions

F1: (Domain/Range) Find the domain and range of each:

(a) $f(x) = \sqrt{4 - x^2}$

(b) $g(x) = \dfrac{1}{x-3}$

(c) $h(x) = \sqrt{x-2} + \dfrac{1}{x-5}$

F2: Is $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = \sqrt{x}$ a function? Explain carefully.

F3: Consider $f : A \to B$ with $A = \{1, 2, 3, 4\}$, $B = \{a, b, c, d\}$, and $f = \{(1, a), (2, b), (3, a), (4, c)\}$. Is f a function? Why?

F4: Let $f : \mathbb{R} \to \mathbb{R}$ be $f(x) = 3x + 2$. Show f is bijective and find f^{-1}.

F5: Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$. (i) Is f injective? (ii) Is f surjective? (iii) Modify domain/codomain to make f bijective.

F6: (Optional) Define $f : \mathbb{N} \to \mathbb{N}$ by $f(n) = n+1$. Determine whether f is injective and/or surjective. Justify.

F7: (Optional) Let $f : A \to B$ and $g : B \to C$. Prove that if both f and g are injective, then $g \circ f$ is injective. Does the analogous statement hold for surjectivity? Discuss.

F8: Let $f : \mathbb{R} \setminus \{1\} \to \mathbb{R} \setminus \{2\}$ be $f(x) = \dfrac{2x+1}{x-1}$. Show that f is bijective and find $f^{-1}(x)$.